Digital Transformation and Sectoral Transition

FinTech, Machine Learning, and Structural Change in Transition Economies

This note formalizes how FinTech (F) and AI/ML (G) drive the movement of labor across sectors. The organizing idea: G **changes *where* labor wants to go;** F **changes *how fast* it can get there.** Each block gives the equation and the meaning of its parameters.

1. The object: structural change

Let $s \in \{a, m, \tau, k\}$ index agriculture, manufacturing/industry, traditional services, and knowledge-intensive services. Let θ_{ist} be sector s's employment share in country i ($\sum_s \theta_{ist} = 1$) and p_{ist} its labor productivity. Aggregate productivity is

$$P_{it} = \sum_s \theta_{ist}\, p_{ist}. \tag{1}$$

The McMillan–Rodrik decomposition splits its growth into a within- and a between-sector part:

$$\Delta P_{it} = \underbrace{\sum_s \theta_{is,t-1}\, \Delta p_{ist}}_{\text{within-sector growth}} + \underbrace{\sum_s p_{is,t-1}\, \Delta \theta_{ist}}_{\text{between-sector (structural change)}} . \tag{2}$$

- Sectoral transition is the *between* term: labor reallocating toward higher-productivity sectors.
- F and G act on *both* terms, through different channels developed below.

2. Sector-specific productivity (where AI bites unevenly)

$$p_{ist} = p_{is,t-1}\Big(1 + \alpha_s^F F_t + \alpha_s^G G_t + \alpha_s^{FG} F_t G_t - \alpha_s^D \rho_s G_t\Big). \tag{3}$$

- α_s^G — AI's productivity effect in sector s, with the central restriction $\alpha_k^G > \alpha_\tau^G > \alpha_m^G > \alpha_a^G \geq 0$ (gains concentrate in knowledge-intensive services).
- α_s^F — FinTech's productivity effect via capital deepening (credit reaching firms in s).
- α_s^{FG} — joint-digital complementarity within the sector, > 0.
- ρ_s — routine-task exposure of sector s (high in m and clerical τ); $\alpha_s^D > 0$.
- $\alpha_s^D \rho_s G_t$ — displacement drag, largest where work is routinizable.

3. Where labor wants to go: the target share

Allocation is drawn toward sectors with higher productivity-adjusted returns (a Roy/logit rule):

$$\theta_{ist}^* = \frac{\exp\big(\eta \, \ln p_{ist}\big)}{\sum_r \exp\big(\eta \, \ln p_{irt}\big)}. \tag{4}$$

- $\eta > 0$ — responsiveness of the desired allocation to relative sectoral returns.
- θ_{ist}^* — frictionless "target" share; a rise in G raises p_{ikt} relative to others and so lifts θ_{ikt}^* (AI tilts the desired structure toward KIS), while displacement lowers p_{imt} and pulls θ_{imt}^* down.

4. How fast it gets there: the FinTech-driven adjustment speed

Reallocation is sluggish; expanding sectors need finance for entry and capital. FinTech relaxes that friction:

$$\Delta\theta_{ist} = \phi(F_t, G_t)\left(\theta^*_{ist} - \theta_{is,t-1}\right), \tag{5}$$

$$\phi(F_t, G_t) = \phi_0 + \phi_F F_t + \phi_G G_t + \phi_{FG} F_t G_t, \qquad \phi(\cdot) \in [0,1]. \tag{6}$$

- ϕ_0 — baseline reallocation speed absent digital technology.
- $\phi_F > 0$ — FinTech accelerates reallocation: credit lets workers and firms chase the new high-return sector. *The core FinTech transition channel.*
- $\phi_{FG} > 0$ — joint digital accelerates it further (algorithmic credit funds exactly the firms AI is making productive).

5. Net structural change

Substituting (5) into the between-sector term:

$$\sum_s p_{is,t-1}\,\Delta\theta_{ist} = \phi(F_t, G_t)\sum_s p_{is,t-1}\big(\theta^*_{ist} - \theta_{is,t-1}\big). \tag{7}$$

- Sign depends on whether the target gap points toward high-productivity KIS (*growth-enhancing* transformation) or toward low-productivity informal sectors absorbing displaced workers (*growth-reducing*).
- For low-H, low-σ^{KIS} economies: θ^*_k barely rises (no KIS base) while θ^*_m falls, so labor pools into traditional/informal services — fast reallocation in the *wrong* direction. This is premature deindustrialization, microfounded.

6. The two opposing forces, made explicit

Differentiating the desired KIS share with respect to AI:

$$\frac{\partial\theta^*_{ikt}}{\partial G_t} \;\propto\; \underbrace{\eta\,\alpha^G_k}_{\text{pull into KIS}} \;-\; \underbrace{\eta\sum_{s\neq k}\theta^*_{ist}\,\alpha^D_s\,\rho_s}_{\text{push out of routine sectors}} \quad . \tag{8}$$

- Productive transformation requires the *pull* (KIS augmentation) to beat the *push* (routine displacement).
- The pull is large only with a pre-existing KIS base and complementary human capital — the absorptive-capacity condition, recast in sectoral terms.

7. Empirical specifications

Construct the between-sector reallocation component $\text{BET}_{it} = \sum_s p_{is,t-1}\Delta\theta_{ist}$, then estimate:

(7a) Reallocation speed — tests $\phi_F, \phi_{FG} > 0$.

$$\text{BET}_{it} = a + \pi_1 F_{it-1} + \pi_2 G_{it-1} + \pi_3 \big(F_{it-1} G_{it-1}\big) + \Lambda' X_{it-1} + \mu_i + \lambda_t + u_{it}. \tag{9}$$

Prediction: $\pi_1 > 0$, $\pi_3 > 0$ (FinTech, and joint digital, speed growth-enhancing reallocation).

(7b) Direction of transformation — tests the ordering $\alpha_k^G > \alpha_m^G$. Sector-specific share-change regressions, run for each s:

$$\Delta\theta_{ist} = a_s + b_s^G G_{it-1} + b_s^F F_{it-1} + b_s^{FG}\big(F_{it-1} G_{it-1}\big) + c_s\big(\rho_s G_{it-1}\big) + \Lambda_s' X_{it-1} + \mu_i + \lambda_t + e_{ist}. \tag{10}$$

Predictions: $b_k^G > 0$ (AI grows KIS employment) and $b_m^G < 0$ or $c_m < 0$ (AI shrinks/displaces manufacturing employment).

(7c) Mediation to growth — does transformation pay off?

$$g_{it} = b + \beta_F F_{it-1} + \beta_G G_{it-1} + \beta_{FG}\big(F_{it-1} G_{it-1}\big) + \vartheta\, \text{BET}_{it-1} + \Psi' X_{it-1} + \mu_i + \lambda_t + \varepsilon_{it}. \tag{11}$$

- If $\pi_3 > 0$ and $\vartheta > 0$: joint digital drives growth-enhancing reallocation.
- If $b_m^G < 0$ is strong while b_k^G is weak for low-H countries: direct evidence of AI-driven premature deindustrialization in transition economies.

Data. Sectoral employment *and* productivity by sector require GGDC 10-Sector or ILOSTAT (ISIC sections), the same source used for the original McMillan–Rodrik decomposition.